THE NAIVE COTANGENT COMPLEX OF HENSELIZATION AND STRICT HENSELIZATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The relative naive cotangent complex of a filtered colimit of etale algebras is acyclic. Algebra, Lemma \ref{algebra-lemma-colimits-NL} and Algebra, Definition \ref{algebra-definition-etale}, cited at proof lines 8536-8539. Makes the intermediate naive cotangent complexes vanish on cohomology.
- The Jacobi-Zariski sequence for a chain of ring maps. Algebra, Lemma \ref{algebra-lemma-exact-sequence-NL}, cited at proof lines 8540-8542. Removes the henselization of the source from the comparison.
- An etale ring map is a global complete intersection, and the six-term Jacobi-Zariski sequence for a colimit of local complete intersections. Algebra, Lemma \ref{algebra-lemma-etale-standard-smooth} and Lemma \ref{lemma-transitive-colimit-lci-at-end}, cited at proof lines 8545-8548. Produces the exact sequence that identifies the base change of the naive cotangent complex with the one over the henselization.

Source reference. Original lemma 0D08: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16303–16303. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 04GS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5843–5843. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 04GU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5845–5845. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07BQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9350–9350. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00U1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1095–1095. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00S2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1024–1024. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00U9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1103–1103. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07D5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9400–9400. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $A \to B$ be a local homomorphism of local rings. Let $A^h \to B^h$, resp. $A^{sh} \to B^{sh}$ be the induced map on henselizations, resp. strict henselizations (Algebra, Lemma 04GS, resp. Lemma 04GU). Then $NL_{B/A} \otimes_B B^h \to NL_{B^h/A^h}$ and $NL_{B/A} \otimes_B B^{sh} \to NL_{B^{sh}/A^{sh}}$ induce isomorphisms on cohomology groups.*

Proof. Since A^h is a filtered colimit of étale algebras over A we see that $NL_{A^h/A}$ is an acyclic complex by Algebra, Lemma 07BQ and Algebra, Definition 00U1. The same is true for B^h/B. Using the Jacobi-Zariski sequence (Algebra, Lemma 00S2) for $A \to A^h \to B^h$ we find that $NL_{B^h/A} \to NL_{B^h/A^h}$ induces isomorphisms on cohomology groups. Moreover, an étale ring map is a local complete intersection as it is even a global complete intersection, see Algebra, Lemma 00U9. By Lemma 07D5 we get a six term exact Jacobi-Zariski sequence associated to $A \to B \to B^h$ which proves that $NL_{B/A} \otimes_B B^h \to NL_{B^h/A}$ induces isomorphisms on cohomology groups. This finishes the proof in the case of the map on henselizations. The case of strict henselization is proved in exactly the same manner. □